

Micro-architectural signature of the service CPU — work OUTSIDE inference
(run_20260609, all tiers)

CPU-seconds (total)	808	86	19	3	1	5
IPC	0.71	1.17	1.97	1.28	0.76	0.91
Branch-miss %	0.84	1.67	1.88	1.31	2.01	2.24
LLC cache-miss %	15.33	21.12	22.86	20.65	30.21	22.13
Mem-bound stall %	5.17	4.76	7.45	6.76	14.72	9.63
Total stall %	70.98	59.70	42.69	55.88	70.61	61.82
dTLB MPKI	0.25	0.46	0.18	0.29	0.62	0.37
MLP (L1 pend/cyc)	4.46	1.64	2.09	1.63	1.60	1.61
	FastAPI (orchestr.)	Milvus (RAG+cache)	MongoDB (cache)	SeaweedFS filer	SeaweedFS volume	Ilm-d gateway

Colour = per-metric min-max across pods (light = low, dark = high vs the other pods); numbers are true values.